Day 042 | Unit 03 | Teacher plan

Integers and Rational Numbers

Reflect Points Across Axes

Standards: 6.NS.C.6b

Learning target: I can reflect points across an axis using signs.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1

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01 / Learn

Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis.

Worked example 1

Plot (-2, -7), reflect it across y-axis, and give the reflected coordinates.

1. Across the x-axis, keep x and reverse the sign of y.

2. Across the y-axis, reverse x and keep y.

3. Reverse the x sign: (2, -7).

4. State the result in the form requested: (2, -7).

Answer: (2, -7)

Worked example 2

Point P is (-4, 3). Select all correct reflections.
A. Across both axes: (-4, 3).
B. Across the x-axis: (-4, -3).
C. Across the x-axis: (4, 3).
D. Across the y-axis: (4, 3).

1. Across the x-axis, keep x and reverse the sign of y.

2. The x-axis reflection changes y only

3. the y-axis reflection changes x only.

4. Across both gives (4, -3).

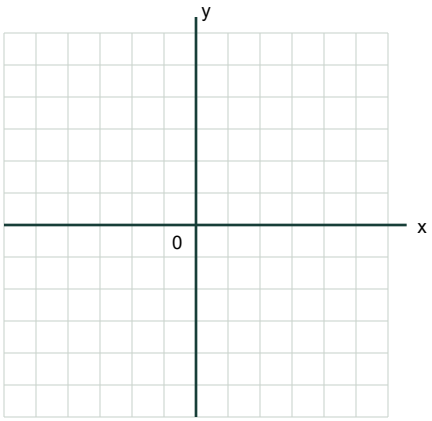
Answer: B, D - Across the x-axis: (-4, -3).; Across the y-axis: (4, 3).

Day 042 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

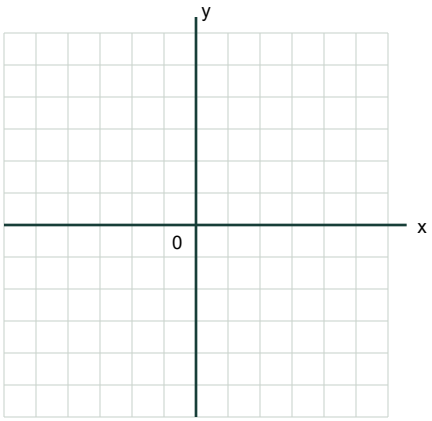
Plot (-3, 4), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

02 / Constructed response

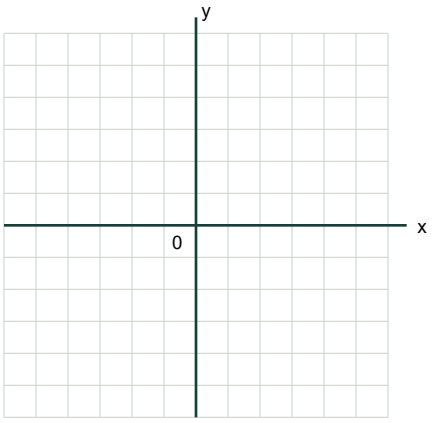
Plot (-4, -7), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

03 / Constructed response

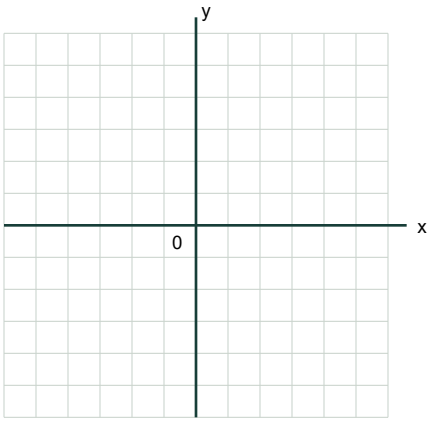
Plot (-8, 6), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

04 / Constructed response

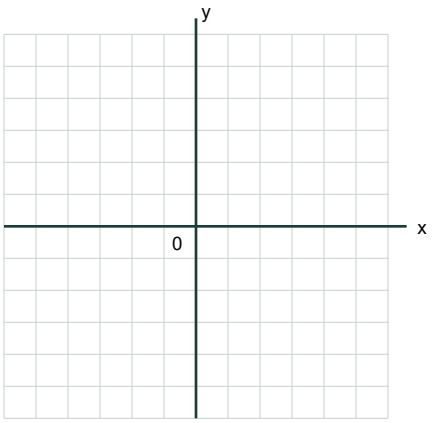
Plot (-7, -7), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

05 / Constructed response

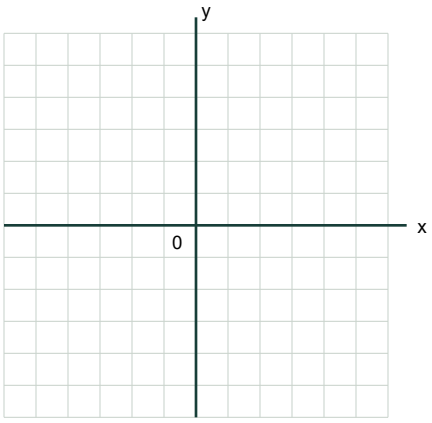
Plot (-4, 3), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

06 / Constructed response

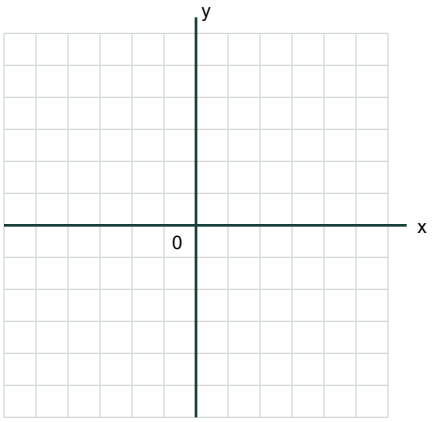
Plot (-7, -3), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

07 / Constructed response

Plot (-8, 4), reflect it across both axes, and give the reflected coordinates.



Label the scale and units.

08 / Multiple select

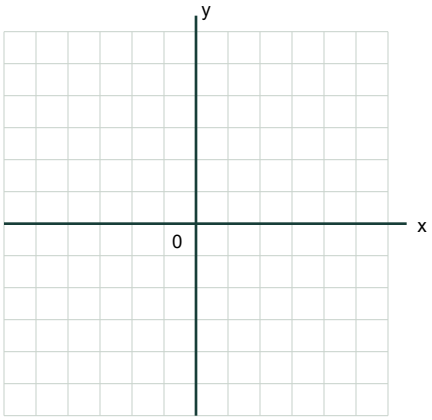
Point P is (-6, 4). Select all correct reflections.
A. Across the y-axis: (6, 4).
B. Across the x-axis: (6, 4).
C. Across the x-axis: (-6, -4).
D. Across both axes: (-6, 4).

09 / Multiple select

Point P is (-2, 2). Select all correct reflections.
A. Across the x-axis: (2, 2).
B. Across the x-axis: (-2, -2).
C. Across the y-axis: (2, 2).
D. Across both axes: (-2, 2).

10 / Constructed response

Plot (-8, -4), reflect it across both axes, and give the reflected coordinates.



Label the scale and units.

Day 042 | Unit 03 | Answer key

Integers and Rational Numbers

Reflect Points Across Axes

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1. (-3, -4)

Reverse the y sign: (-3, -4).

6.NS.C.6b | Constructed response

2. (4, -7)

Reverse the x sign: (4, -7).

6.NS.C.6b | Constructed response

3. (-8, -6)

Reverse the y sign: (-8, -6).

6.NS.C.6b | Constructed response

4. (7, -7)

Reverse the x sign: (7, -7).

6.NS.C.6b | Constructed response

5. (-4, -3)

Reverse the y sign: (-4, -3).

6.NS.C.6b | Constructed response

6. (7, -3)

Reverse the x sign: (7, -3).

6.NS.C.6b | Constructed response

7. (8, -4)

Reverse both signs: (8, -4).

6.NS.C.6b | Constructed response

8. A, C - Across the y-axis: (6, 4).; Across the x-axis: (-6, -4).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (6, -4).

6.NS.C.6b | Multiple select

9. B, C - Across the x-axis: (-2, -2).; Across the y-axis: (2, 2).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (2, -2).

6.NS.C.6b | Multiple select

10. (8, 4)

Reverse both signs: (8, 4).

6.NS.C.6b | Constructed response